

Model-Informed Evaluation of Hydroxyurea Exposure During Lactation

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Hydroxyurea is a cornerstone therapy for sickle cell anemia; however, evidence guiding its use during lactation remains limited. This study aimed to develop a population pharmacokinetic (PK) model to characterize hydroxyurea disposition in maternal plasma and breast milk, and to quantify infant exposure under clinically relevant breastfeeding scenarios. Population PK modeling was conducted using data from the HELPS study (NCT02990598) with nonlinear mixed-effects modeling. Maternal plasma PK parameters were first estimated using observed plasma concentration data, followed by estimation of parameters characterizing drug transfer into breast milk. Model-based stochastic simulations were performed across infant ages 0–9 months to evaluate exposure, expressed as the World Health Organization (WHO)-recommended relative infant dose (RID), with 10% as the consensus safety threshold. The model diagnostics demonstrated that both plasma and breast milk PK were adequately described by the model. Simulation analyses demonstrated that the estimated RID remained below 10% across all evaluated infant age groups. This integrated plasma-breast milk population PK model provides a quantitative framework to assess lactational drug exposure. The findings suggest that infant exposure to hydroxyurea through breastfeeding is unlikely to exceed the 10% safety threshold across all age groups (0–9 months), supporting continued maternal therapy during lactation with appropriate clinical oversight.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ The HELPS study demonstrated that estimated hydroxyurea exposure in breastfed infants was below the WHO-recommended 10% relative infant dose (RID) threshold under standard lactation assumptions.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ Can a population PK model incorporating a breast milk compartment quantify variability in infant hydroxyurea exposure across clinically relevant lactation scenarios?

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ Model-informed simulations demonstrate that infant exposure remains below the 10% RID threshold across age groups and feeding patterns, even under conservative assumptions.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ These findings provide quantitative support for evidence-based decision-making regarding continuation of hydroxyurea therapy during lactation in women with SCA.

Sickle cell anemia (SCA) is a life-threatening inherited blood disorder affecting approximately 100,000 individuals in the United States and millions worldwide.^{1–3} It is associated with severe clinical complications, including multi-organ failure, vaso-occlusive pain crisis, stroke, and premature mortality.^{1,2,4} Advances in medical care have substantially improved survival, increasing life expectancy from approximately 10 years in the 1960s to over 45 years in the 2000s.^{1,5,6} With improved survival, SCA has evolved into a chronic condition of adulthood, and parenthood has become an attainable life goal for many patients.¹ Consequently, reproductive

health has emerged as an essential component of comprehensive SCA care.¹

Breastfeeding is the normative standard for infant feeding and provides a balanced source of nutrients, as well as immunologic and bioactive compounds that support growth, immune protection, and neurodevelopment.^{7–9} In contrast, infant formula does not contain many of these human milk-derived bioactive factors and, compared with breastfeeding, has been associated with higher infectious morbidity and adverse long-term health outcomes across some studies.¹⁰ The World Health Organization (WHO) and the American

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Academy of Pediatrics (AAP) recommend exclusive breastfeeding for approximately the first 6 months of life,^{11,12} followed by continued breastfeeding with appropriate complementary foods for up to 2 years of age or longer, as mutually desired.^{7–9,11,12} In regions with a high prevalence of SCA, including Sub-Saharan Africa, where approximately three-quarters of affected births occur, breastfeeding

represents an accessible nutritional strategy, particularly in settings affected by food insecurity. Food insecurity remains a major public health concern globally.^{1,12}

Hydroxyurea is the primary disease-modifying therapy for SCA and is approved by both the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA).^{1,2,11,13}

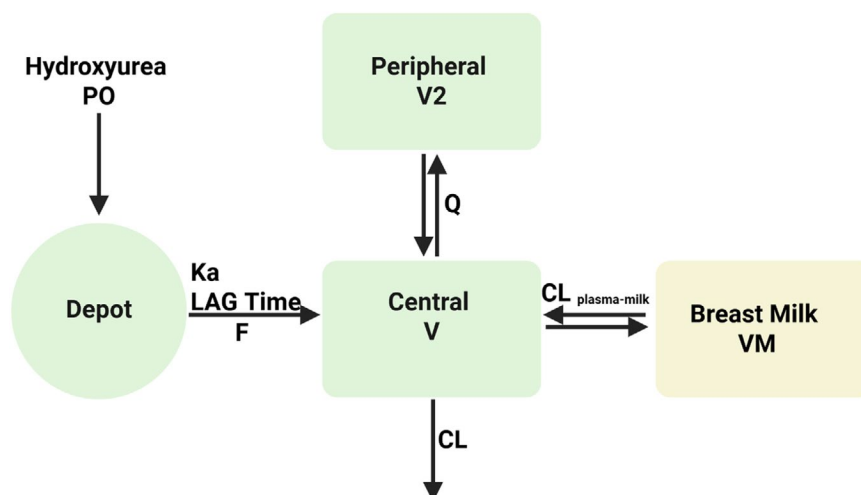


Figure 1 Schematic diagram of the final model structure of hydroxyurea, orally taken (PO). CL, Clearance; V, volume of distribution of the central compartment; Q, internal-compartmental clearance; V2, volume distribution of the peripheral compartment; KA, first-order absorption constant; LAG time, lag time; VM, milk volume; CL plasma-milk, inter-compartmental clearance between plasma and milk compartments.

Table 1 Parameter estimates from the final model and bootstrap analysis

Parameter (Unit)	Estimates	RSE%	Median	95% CI
CL (L/h/70 kg)	8.57	8.65	8.39	6.47–9.65
V(L/70 kg)	39.0	11.5	37.8	31.88–45.18
Q (L/h)	4.4	18.3	4.4	3.39–8.30
V2 (L)	19.1	19.8	19.4	14.59–48.13
KA (/h)	2.5	36.3	2.3	1.43–4.07
LAG Time (h)	0.2	14.2	0.2	0.16–0.24
F	1 Fixed	–	–	–
CL plasma-milk (L/h)	2.3	17.8	2.3	1.71–3.15
VM (L)	0.79	6.80	0.79	0.70–0.86
Interindividual variability (CV%)				
IIVCL	27.4	49.0	25.0	14.5–34.6
IIVV	42.3	30.3	40.5	26.9–53.2
IIVKA	114	35.8	110.9	70.1–197.8
IIVLAG time	43.7	49.1	40.1	15.4–60.5
IIVCL plasma-milk	51.7	49.8	48.0	24.7–78.9
IIVVM	23.2	45.4	23.2	11.9–30.1
Residual variability				
Prop.Err.(CV%)	15.0	28.9	15.2	10.9–18.7
Add.Err. (µg/mL)	1.12	34.0	1.10	0.53–1.71

CL, clearance; V and volume of distribution are allometrically size-standardized with a power of 0.75 for clearance and 1.00 for volume of distribution. Q, intercompartmental clearance; V2, volume distribution of the peripheral compartment; KA, first-order absorption constant; LAG time, lag time; CL plasma-milk, intercompartmental clearance between plasma and milk compartments; VM, milk volume; RSE, relative standard error; CI, confidence interval; IIV, interindividual variability; CV, coefficient of variation. Prop.Err, proportional part of the residual unexplained variability for hydroxyurea plasma concentrations. Add.Err, additive part of the residual unexplained variability for hydroxyurea milk concentrations.

Extensive clinical evidence over the past 25 years has established its safety and efficacy, leading to recommendations for initiation as early as 9 months of age.^{4,11,14} By increasing fetal hemoglobin (HbF) production and reducing inflammation, hydroxyurea reduces vaso-occlusive events, hospitalizations, and mortality.^{1,13,15} Despite these benefits, concerns regarding potential fetal harm based on animal studies have limited its use during pregnancy.¹⁶ It is often discontinued during lactation because of theoretical safety concerns and limited data on infant exposure through breast milk.^{1,11,13} However, withholding hydroxyurea may increase morbidity risk in lactating women with SCA.¹¹

The HELPS study (NCT02990598) reported that estimated infant exposure remained below the WHO-recommended 10% relative infant dose (RID) threshold for safety under standard assumptions of milk intake.¹¹ However, the extent to which variability in breastfeeding patterns, milk intake, and dosing timing may influence infant exposure has not been systematically evaluated. To address this gap, we developed a population PK model incorporating an integrated breastmilk compartment to characterize the hydroxyurea concentration–time profiles in maternal plasma and breast milk. This model enables quantitative assessment of how variations in breastfeeding frequency and the timing of drug administration influence infant exposure. By extending beyond fixed assumptions, this model-informed framework supports evidence-based decision-making regarding continuation of hydroxyurea therapy during lactation.

METHODS

Patients and sampling

Data from the prospective Hydroxyurea Exposure in Lactation Study (HELPS; NCT02990598) were used for model development.¹¹ The HELPS study protocol was approved by the Institutional Review Board. A total of 101 plasma and 93 breast milk hydroxyurea concentrations were obtained from 16 lactating women who received a single oral dose of 1000 mg hydroxyurea (approximately 15–20 mg/kg). Baseline plasma and breast milk samples were collected prior to dosing, followed by serial post-dose sampling. Seven subjects provided plasma samples at 0.25, 0.5, 1, 2, 3, 4, 6, and 8 hours after dosing. Milk samples from these subjects were collected at the same timepoints, with an additional sample at 12 hours. Five subjects provided both plasma and milk samples at 0.25, 0.5, 1, 2, 3, 4, 6, 8, 12, and 25 hours after dosing. Two subjects provided plasma and milk samples at 3, 6, 9, and 25 hours after dosing. For the remaining two subjects, only milk samples were collected. Sampling times were 3, 6, 8, and 24 hours for one subject, and 1, 2, 3, 6, 9, and 24 hours for the other. Hydroxyurea concentrations in plasma and breast milk were measured using validated high-performance liquid chromatography methods as previously described.¹¹

Population PK modeling

Population PK analysis was conducted using nonlinear mixed-effects modeling with NONMEM version 7.5.0 (ICON Development Solutions, Ellicott City, MD, USA). First-order conditional estimation with interaction (FOCE-I) was employed to estimate structural PK parameters, inter-individual variability (IIV), and residual variability (RV).

Plasma concentrations were modeled first with compartmental models to characterize systemic disposition. Lag-time and transit-compartment models were evaluated to describe potential delay in hydroxyurea absorption. A breast milk compartment was subsequently integrated into the

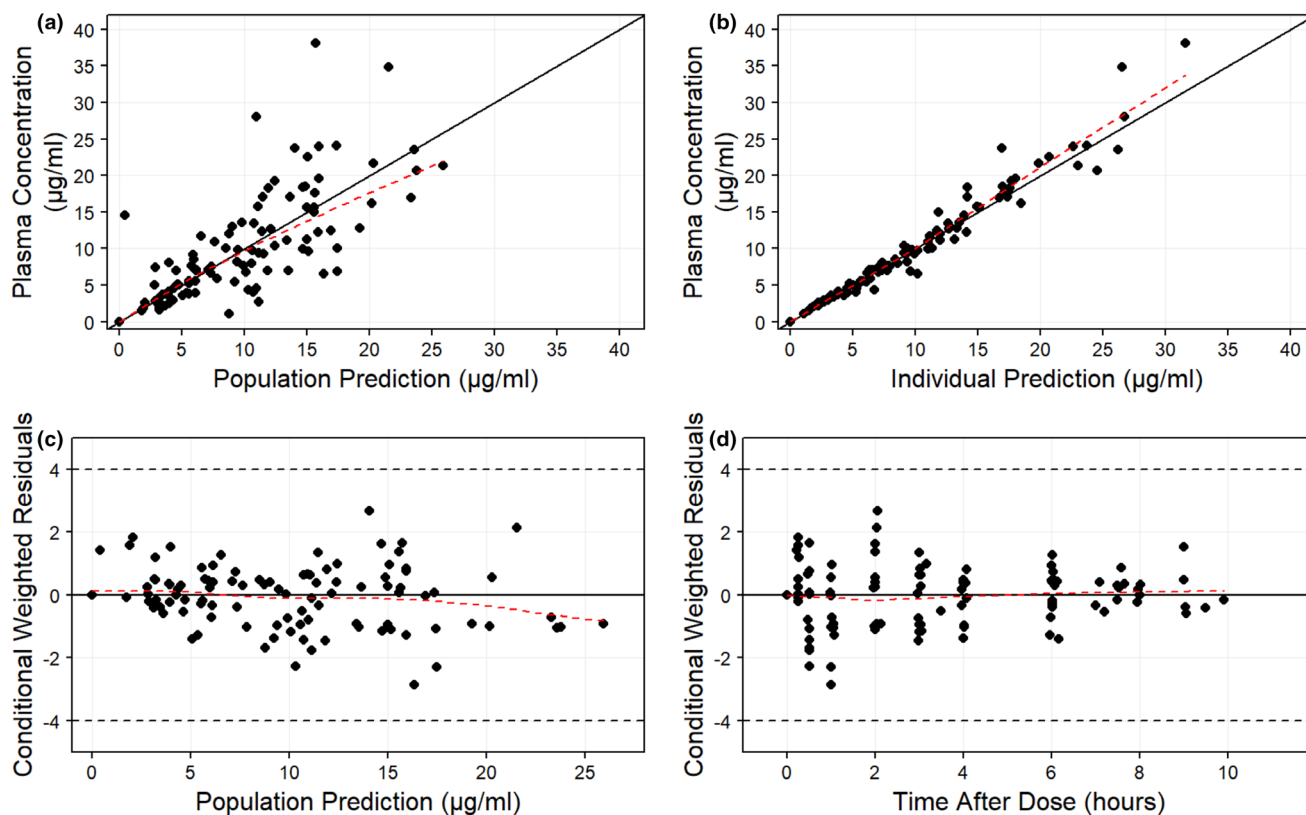


Figure 2 Goodness-of-fit plots for the final model (Plasma). (a) Population prediction vs. observations. (b) Individual prediction vs. observations. (c) Conditional weighted residuals vs. population prediction. (d) Conditional weighted residuals vs. time after dose. Solid line, unity line. Red line, local regression line.

plasma model to characterize drug transfer into milk, estimating milk volume and inter-compartmental clearance (Figure 1).

IIV was described using an exponential error model as follows (Eq. 1)

$$P_i = P_{\{\text{pop}\}} \times \exp(\eta_i) \quad (1)$$

where P_i is the individual parameter estimate, P_{pop} is the typical population value, and η_i is an inter-individual random effect for individual i with the mean of 0 and variance of ω^2 . Residual variability was evaluated using proportional, additive, or combined error models.

Coefficient of variation (CV%) for ω^2 and residual error (σ^2) was calculated from the square roots of their respective variances, with ω^2 converted to CV% using the standard log-normal formula as follows¹⁷:

$$\text{CV\%} = 100 \times \sqrt{\exp(\omega^2) - 1} \quad (2)$$

Covariate analysis

Body weight was evaluated as a covariate using allometric scaling, with theoretical exponents of 0.75 for apparent clearance (CL) and inter-compartmental clearance (Q), and 1 for volumes of distribution of the central compartment (V) and peripheral compartment (V2), standardized to 70 kg. A reduction in the objective function value (OFV) of 3.84 ($P < 0.05$) was considered statistically significant to include body weight in the final model.

Model evaluation

Model selection was based on convergence, parameter plausibility, and goodness-of-fit diagnostics. Parameter precision and stability were

assessed using nonparametric bootstrap analysis (1000 replicates). The final model was considered stable if bootstrap median parameter estimates were within 10% of the final estimates and 95% confidence intervals excluded zero. Predictive performance was evaluated using visual predictive checks (1000 simulations).

Model-informed simulation analysis

Monte Carlo simulations were performed using NONMEM to evaluate lactation scenarios across infant age groups (0–9 months old). A virtual population of 1000 lactating women was generated using demographic distributions from NHANES to simulate hydroxyurea concentrations in breast milk at a standard dose of 20 mg/kg/day.^{18,19} The dose was assumed to be administered at time zero, and hydroxyurea milk concentrations were predicted over the 24 hours following dose administration. An age range of 18–45 years was selected to represent women of childbearing age with SCA. The body weight range was chosen to be consistent with the HELPS study population, as the participants were breastfeeding mothers and therefore expected to have body weights representative of this group.¹¹

Age-dependent infant's daily milk intake was estimated as follows (Eq. 3, according to a previous publication).²⁰

$$\begin{aligned} \text{Daily milk intake (mL/kg/day)} \\ = 160.39 \times \frac{0.232}{0.232 - 0.00252} \times (e^{-0.00252 \times \text{age}} - e^{-0.232 \times \text{age}}) \end{aligned} \quad (3)$$

Feeding frequency varied based on the infant's age, and infants were categorized into age groups based on the number of feedings per day.^{21–23} The age groups and corresponding breastfeeding frequencies were defined as follows: Group 1 (0 to < 1 month), 12 times/day; Group 2 (1 to < 2 months), 9 times/day; Group 3 (2–3 months), 8 times/day; Group 4 (4–6 months), 6 times/day; Group 5 (7–9 months), 4–5 times/day.

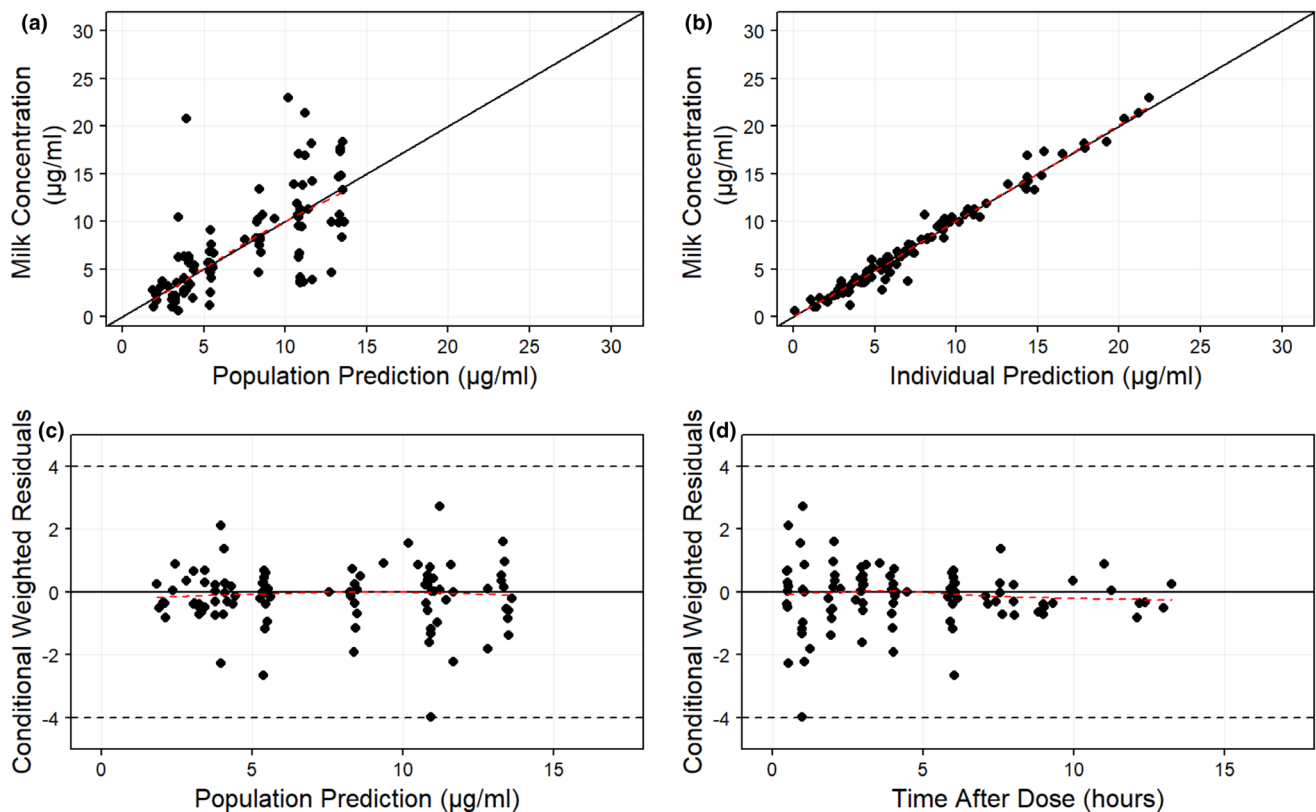


Figure 3 Goodness-of-fit plots for the final model (milk). (a) Population prediction vs. observations. (b) Individual prediction vs. observations. (c) Conditional weighted residuals vs. population prediction. (d) Conditional weighted residuals vs. time after dose. Solid line, unity line. Red line, local regression line.

The hydroxyurea dose per feed (mg/kg) was obtained with the model-predicted milk concentrations at the corresponding feeding times and feed volume (Eq. 4).

$$\text{Dose per feed } (\mu\text{g/kg}) = C_{\text{milk}} (\mu\text{g/mL}) \times V_{\text{feed}} (\text{mL/kg}) \quad (4)$$

The infant's daily dose of hydroxyurea (mg/kg/day) was estimated by summing all hydroxyurea doses per feed (Eq. 5).

$$\text{Infant's daily dose} = \sum \text{Dose per feed} \quad (5)$$

To evaluate safety, the estimated daily infant's dose was compared with the RID thresholds of 10% (2 mg/kg/day of hydroxyurea), consistent with WHO guidance and Hale's Medications & Mothers' Milk.^{24,25}

RESULTS

Population PK modeling

A two-compartment model with first-order absorption and a lag time adequately described the hydroxyurea plasma concentration-time profiles. Body weight was identified as a significant

covariate, and its inclusion significantly improved the model fit ($\Delta\text{OFV} = -11.9, P < 0.001$). The addition of a breast milk compartment enabled the characterization of milk concentration-time profiles. Final population PK parameter estimates are summarized in Table 1. IIV was most pronounced for the absorption rate constant (coefficient of variation, 114%). Residual variability was best described by a proportional error model for plasma concentrations and an additive error model for milk concentrations.

Goodness-of-fit plots demonstrated an overall reasonable model fit (Figures 2 and 3). The nonparametric bootstrap analysis showed the stability and reliability of the final model parameter estimates (Table 1). VPC analyses showed a reasonable agreement between the simulated and observed concentrations (Figure 4). Additionally, the model captured the individual PK profiles in both maternal plasma and breast milk.

Model-informed simulations

Simulated milk intake increased from birth, reaching a peak of approximately 152.6 mL/kg/day at 19.7 days of age, followed by

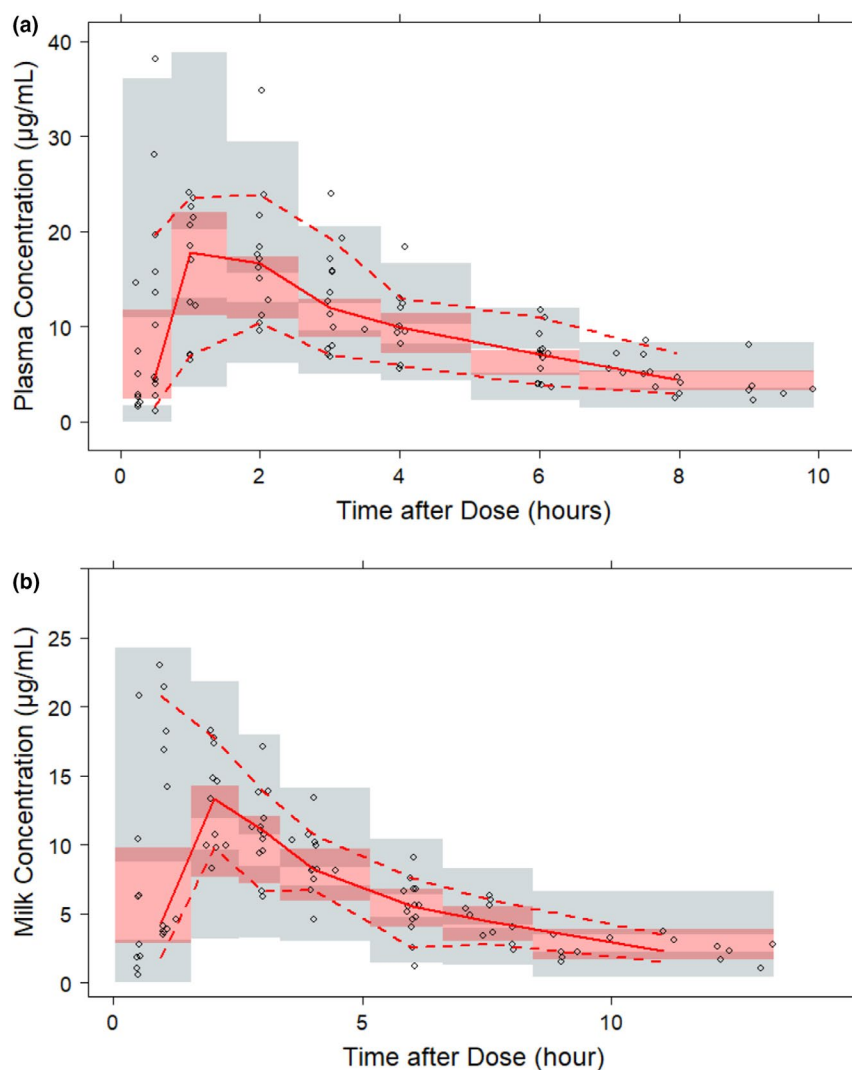


Figure 4 Visual predictive check of the final model (a: plasma, b: milk). Circles, observed plasma concentrations; red dashed lines, observed 10th and 90th percentiles; red solid line, observed median; shaded areas, 95% confidence intervals around the percentile predictions.

a gradual decline with increasing infant age. Predicted infant hydroxyurea exposure mirrored this trajectory. Estimated infant exposure increased during the early postnatal period in parallel with rising milk intake, peaked (median: 0.76 mg/kg/day) near the time of maximal milk consumption, and subsequently declined as milk intake and feeding frequency decreased (Figure 5 and Table 2).

Across all evaluated age cohorts (0–9 months), the RID remained below the 10% safety threshold. This indicates that, regardless of the infant's age, the estimated exposure through breast milk represents less than 10% of the maternal dose.

DISCUSSION

This study presents the first population PK model incorporating a dedicated breast milk compartment to characterize hydroxyurea PK during lactation. The model accounted for inter-individual variabilities in hydroxyurea PK parameters among lactating women. Body weight was identified as a significant covariate influencing clearance and volume of distribution, consistent with previously published population PK models.^{4,26,27}

The estimated apparent CL (8.57 L/h/70 kg) is consistent with the previously reported oral clearance of 73.16 mL/min/m² by Rodriguez *et al.*, corresponding to approximately 7.6–7.9 L/h for a typical adult body surface area (1.73–1.8 m²).²⁸ We therefore interpret our CL estimate as physiologically plausible and within the range of published values, while acknowledging that inclusion

of renal function markers could further refine the inter-individual variability in the clearance estimate.

Hydroxyurea has a reported volume of distribution that approximates total body water.²⁹ A population PK study in patients with SCA reported a volume of distribution of 45.3 L/70 kg.²⁶ In addition, a population PK study in cancer patients described the volume of distribution as $V = 0.186 \times (\text{body weight}) + 25.4 \text{ L}$, corresponding to approximately 38.4 L/70 kg.³⁰ The volume of distribution estimated in our model (39 L/70 kg) is consistent with previously reported values.

Our model-estimated K_a (2.5 h⁻¹) falls within the reported range of approximately 2–3 h⁻¹, indicating rapid oral absorption.^{26,30}

Our model indicated rapid equilibrium between the plasma and milk compartments (CL plasma-milk = 2.26 L/h), consistent with what was reported in the HELPS study.¹¹

The model was utilized to estimate infant exposure to hydroxyurea through simulations of clinically relevant lactation scenarios. In our simulations, daily milk intake and feeding frequency were informed by the weight-normalized human milk intake (WHMI) curve reported by Yeung *et al.*^{20,31} This model predicts peak milk intake of approximately 152.6 mL/kg/day at around 20 days of age, followed by a gradual decline, reflecting early postnatal feeding physiology. Epidemiological data further suggest that milk volume increases during early infancy, followed by a plateau through approximately 6 months and subsequent reductions as

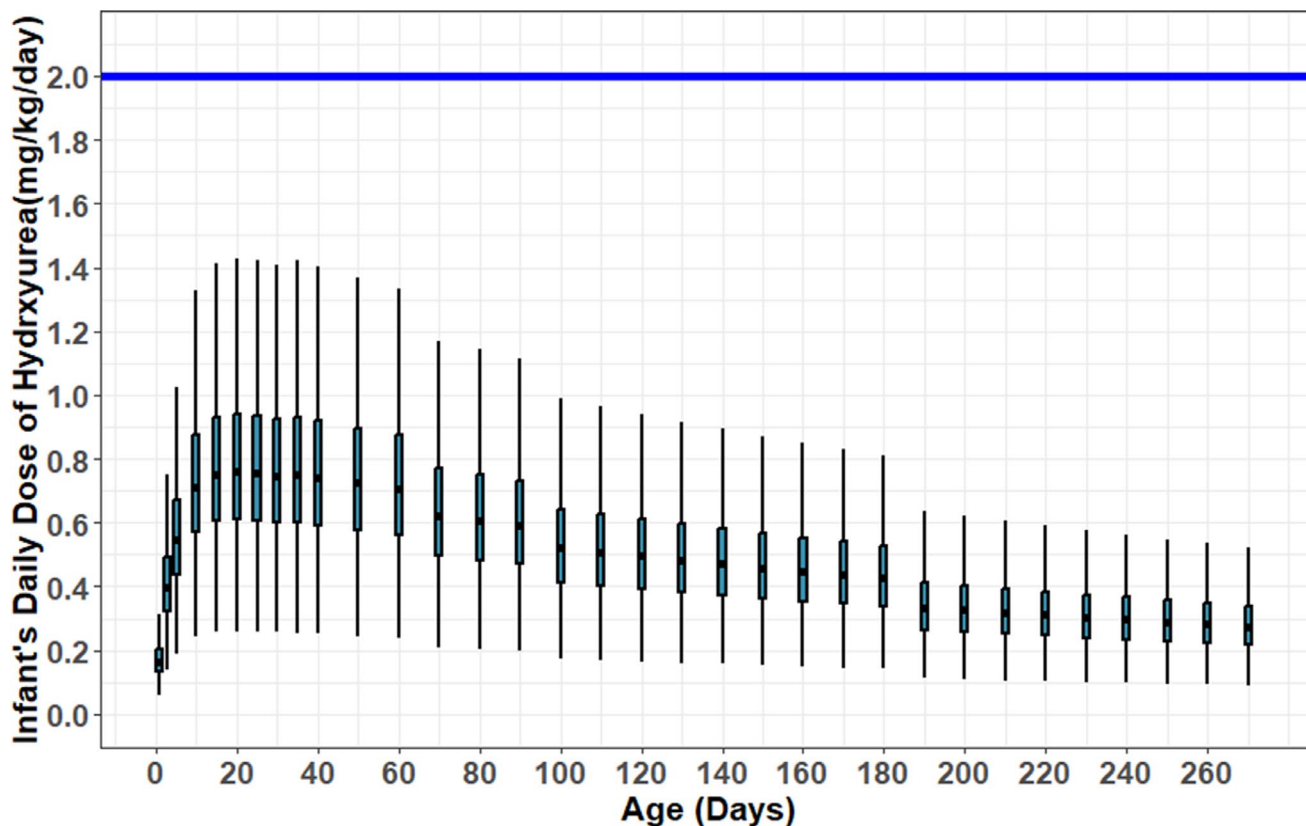


Figure 5 Infant's daily dose of hydroxyurea via breastfeeding. Data are binned based on milk intake and predicted hydroxyurea dose. The vertical line: the overall range of HU_intake at that specific age. The blue box, the interquartile range (IQR: 25th–75th percentile); the horizontal line inside the box, the median HU_intake; the top whisker, maximum HU_intake; the bottom whisker, minimum HU_intake.

Table 2 Infant's daily milk intake and hydroxyurea dose

Age (months)	Milk intake (mL/kg/day)	Feeding times (/day)	Hydroxyurea dose (mg/kg/day)	Hydroxyurea dose (mg/kg/day)
			Median	5 th –95 th percentile
1	150.3	12	0.75	0.44–1.30
2	139.6	9	0.71	0.41–1.23
3	129.5	8	0.59	0.34–1.03
4	120.1	6	0.49	0.28–0.87
5	111.4	6	0.46	0.26–0.80
6	103.4	6	0.43	0.24–0.75
7	95.9	4	0.32	0.18–0.56
8	89.0	4	0.30	0.17–0.52
9	82.6	4	0.27	0.16–0.48

Milk intake, feeding times, and hydroxyurea dose per day are shown for each age cohort (1–9 months). Milk intake is presented as mL/kg/day, and hydroxyurea dose is presented as median (5th–95th percentiles), mg/kg/day.

complementary feeding is introduced.³² The simulated milk intake trajectories fell within these reported epidemiologic patterns.

Simulation results showed that the RID of infant's hydroxyurea dose (median range: 0.27–0.76 mg/kg/day) remained below the generally accepted safety threshold of 10% even under worst-case conditions. These findings are consistent with a recent physiologically based pharmacokinetic (PBPK) modeling study that evaluated hydroxyurea disposition in lactating women and breastfed infants using a mechanistic maternal–infant framework, with predicted RID values remaining below the 10% threshold.³³ The concordance between our PopPK simulations and the PBPK-based predictions supports that hydroxyurea exposure through breastfeeding is low and may be continued during lactation under appropriate clinical supervision and strengthens the evidence base for informed dosing and breastfeeding recommendations in women with SCA receiving hydroxyurea therapy.

Currently, the FDA label for hydroxyurea advises discontinuing breastfeeding. The results of our study suggest that infant exposure through breast milk is below the recommended 10% RID threshold across all infants aged 0–9 months. These findings provide quantitative, model-informed evidence that may support a reevaluation of current labeling. In particular, integrating such data with emerging clinical evidence could inform a more nuanced, risk-based approach to lactation recommendations, rather than defaulting to discontinuation of breastfeeding. From a clinical implementation perspective, one of the major remaining uncertainties is the lack of clinical PK data in breastfed infants, particularly those younger than 6 months of age. Additional data characterizing systemic exposure in breastfed infants would help reduce this uncertainty and support future clinical decision-making and regulatory discussions. Pediatric physiologically based pharmacokinetic (PBPK) modeling may complement these efforts by translating the observed breast milk exposure data into age-specific predictions of infant systemic exposure and helping inform the design and interpretation of future clinical studies.

A key limitation of this study is the absence of renal function data. Because hydroxyurea is primarily eliminated renally and patients with SCA may experience renal impairment, assumptions of normal renal function may have influenced clearance estimates in

some individuals. Future studies incorporating renal biomarkers are warranted to refine exposure predictions and further characterize variability in lactating populations.

CONCLUSION

We developed a population PK model describing hydroxyurea disposition in plasma and breast milk during lactation. Model-informed simulations indicate that infant exposure remains below the 10% relative infant dose safety threshold across all evaluated infant age groups. This framework supports quantitative risk assessment and informed clinical decision-making regarding hydroxyurea therapy during lactation.

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AI-assisted tools were used only for language and formatting improvements. The authors retain full responsibility for the final manuscript.

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CONFLICT OF INTEREST

The authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

AH, KI, MD, JW, RW, and TM wrote the manuscript; AH, KI, MD, and TM designed the research; AH, KI, MD, JW, RW, and TM performed the research; and AH, KI, MD, and TM analyzed the data.

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